

Food Inventory from Egocentric Kitchen Video

ABSTRACT

We target automatic, per-item inventory tracking in home kitchens from first-person video. Given a participant’s current inventory (items with known starting amounts) and a cloud-uploaded kitchen-activity video, the system outputs, for each inventory item, how much was consumed and how much remains, under an objective of *minimizing VLM cost subject to an accuracy requirement*. The core insight is that inventory-relevant interactions occupy only a small fraction of cooking video (roughly 6% in HD-EPIC), so lightweight CV models can pre-select the short observation windows over which an expensive VLM must reason. We are collecting a benchmark of in-the-wild and controlled-consumption recordings across four households with Meta Ray-Ban Smart Glasses, and we evaluate against per-session ground-truth consumption using a capacity-normalized percentage error (CNPE). This draft reports the problem statement, system design, data-collection targets and current progress, and the evaluation plan; introduction, related work, and results are deferred to later revisions.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; • **Human-centered computing** → *Ubiquitous and mobile computing*.

KEYWORDS

egocentric video, food inventory, vision–language models, wearable sensing, active perception

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1 PROBLEM STATEMENT

Given a participant’s current inventory (items with known starting amounts and states) and an untrimmed kitchen-activity video recorded from a first-person viewpoint, **predict**, for each inventory item, the consumed amount and the remaining amount after the activity. Videos are uploaded to a cloud server, and inference is performed off-device.

Objective. Minimize VLM cost subject to an accuracy requirement.

1.1 Motivating observations

- **Videos are long; the relevant window is short.** Updating an item’s inventory only requires observing the brief interaction with that item, not the rest of the session.

- **Amount reasoning is hard and needs a powerful VLM.** Deciding how much of a jar of sauce was used, or how many tomatoes were cut, is a fine-grained visual-reasoning task that weaker models do poorly on.
- **Powerful VLM APIs are expensive.** An hour of video sampled at 1 fps costs roughly \$2.5 in Gemini 3 Pro input tokens alone, and current VLM context windows typically cannot fit more than about 10 minutes of video per call.
- **Most of the video is irrelevant.** In HD-EPIC, interactions with inventory items account for only ~6% of cooking video time; the rest is meal prep, cooking, washing dishes, and similar activity [1].

1.2 Design insight

Use lightweight CV models to *locate* segments that interact with inventory items, so that an expensive VLM only reasons over those short windows. This both reduces token cost and makes it feasible to reason across long-horizon recordings that would otherwise exceed the VLM’s context limit.

2 SYSTEM DESIGN

Important question: Is it reasonable or not to assume the starting

Our pipeline takes an untrimmed egocentric kitchen video and a per-participant *ledger* (the list of purchased food items with package amounts), and produces a per-session update of *how much of each item was consumed*. The pipeline is organized as a sequence of numbered stages (scripts in `system_design/`) with well-defined inputs and outputs so that components can be swapped or ablated.

2.1 Pipeline stages

- (1) **Hand–object interaction (HOI grasp detection).** The Hands23 detector runs on sampled frames across the full video and flags frames in which a hand is in *object-contact* with something it is holding. Only these grasp frames are carried forward; frames with no active grasp are dropped, so downstream stages operate on a sparse set of moments when the participant is actually manipulating an item.
- (2) **Scene tagging (OWLv2).** An OWLv2 open-vocabulary detector labels each grasp frame with a coarse kitchen location (storage, sink, stove, countertop, unknown). These tags give the reasoner the lifecycle cue needed to distinguish retrieval from use from put-back when selecting observation windows.
- (3) **Item identification (SigLIP2 + DINOv2).** The in-contact object crop from each grasp frame is matched against the participant’s *current* inventory using SigLIP2 (zero-shot text prompts from item labels) and DINOv2 (image-to-image matching against reference package photos). The two scores together decide *which ledger item* is being handled in each grasp frame.
- (4) **Amount estimation.** The proposed system predicts *how much remained* (or how much was used) for each touched item at the end of the session. This stage is itself decomposed

into two sub-stages so that the expensive vision-language reasoning is invoked exactly once per session, on a planner-selected frame set:

- 4a. **Frame-budget planner (text-only LLM).** The grasp, identification, and scene evidence from stages 1–3 is collapsed into per-item, HOI-gated temporal segments (a chronological list of on-contact bursts, each tagged with peak identification score and a coarse scene distribution). A text-only LLM reads the session inventory together with this evidence and emits, per inventory item, a decision (skip or observe) and an allocation of a shared frame budget into (i) *journey samples* spread across the item’s bursts and (ii) *dense windows* concentrated on the most informative bursts (with a configurable rule biasing toward late, dispense-phase bursts over early retrieval).
- 4b. **Sweep VLM observer.** A single multi-image VLM call sees exactly the planner-selected frames and returns, per observed item, `{item_confirmed, evidence_frames, amount_remaining}`.

Stages 1–3 correspond to the lightweight “locate relevant windows” path described in Section 1; stage 4a is where a cheap language model reasons over the full session at the symbol level, and stage 4b is where the powerful VLM performs the expensive perceptual reasoning, but only on the planner-allocated frames.

Iterative plan–observe–replan. Stage 4 is structured as a plan–observe–replan loop rather than a single shot. After the round-1 sweep, the planner is re-invoked with (i) the actual timestamp ranges of frames the round-1 observer saw (not the planner’s declared windows) and (ii) the per-item answers and confidence the observer returned. The replanner is allowed to revisit any moment in the session — including frames already inspected — so that its budget is spent on items whose remaining amount was not resolved in round 1. The loop terminates when the planner declares done, a round cap is reached, or a session-level frame budget is exhausted; easy sessions finish in one round at low cost, while hard sessions spend their budget adaptively on the items that remain uncertain.

2.2 Reasoner extensions

Cross-session visual memory. Amount estimation within a single session is fundamentally under-determined when the stock container is only partially visible, or when it never appears in its final state. This extension threads a small *visual memory* across sessions: for each item the reasoner confirms, we persist a single canonical end-of-session frame (the moment the stock container is most clearly visible) together with its estimated remaining amount. On subsequent sessions the reasoner is prompted with this item’s prior snapshots, in chronological order, alongside the current-session frames. The prior *numeric* estimates are withheld to avoid anchoring; only the visual evolution of the package is exposed. The intended effect is that the reasoner can compare the current fill level against the previous fill level directly, rather than re-deriving the remaining amount from scratch each session.

2.3 Baseline

A **whole-video** baseline gives the VLM the entire untrimmed video and asks it to predict remaining amounts end-to-end, with no temporal structure supplied. This is the comparator against which the proposed system’s cost–accuracy gain is measured.

3 DATASET OVERVIEW AND COLLECTION PROGRESS

The dataset is collected from three participants cooking in their own kitchens. Each participant records first-person video using either **Meta Ray-Ban Smart Glasses** (egocentric, fisheye) or a head-mounted **Raspberry Pi Camera Module v3**, depending on hardware availability. For each session we also record, for every item the participant interacts with, the item’s starting amount and ending amount (by weight for bulk items, by count for discrete items such as eggs), which serve as the ground-truth consumption label.

3.1 Collection protocol

We collect two complementary flavours of data with the same recording hardware and the same start/end measurement procedure.

In-the-wild cooking sessions. Participants wear the recording device during their ordinary meal-prep sessions at home, with no instructions on what or how to cook. These sessions cover the realistic target scenario — free-form cooking with multiple items, varied background clutter, and whatever item subset happens to be on the shelf that week — and form the primary evaluation cohort.

Controlled consumption sessions. In a controlled session the participant follows a scripted protocol that consumes a *single* pre-selected item at a time: the item is removed from storage, a known amount is poured/scooped/cut out, and the item is returned. Several items are cycled through in a single recording. We include controlled sessions for two reasons:

- (1) **Food variety.** In-the-wild cooking sessions naturally concentrate on a small set of staples that a given participant happens to cook with; controlled sessions let us deliberately exercise items that are under-represented in free-form cooking (e.g. less-used condiments, opaque-packaged items, items with unusual shape categories), broadening the visual and packaging distribution the evaluation covers.
- (2) **Statistical power.** Controlled sessions produce many cleanly-bounded consumption events per unit recording time, which increases the event count available for paired accuracy comparisons and allows us to detect proposed system-vs.-baseline improvements in VLM amount-estimation accuracy at statistical significance that the wild cohort alone, with its smaller event count, would not reliably support.

3.2 Current collection status

Table 1 reports the dataset as it currently stands against the targets for the full benchmark (4 households, 20 h in-the-wild + 2 h controlled, 400 + 300 consumption events, ~70 distinct items across 4 shape categories). Four participants have begun recording; three

(kailai, giraffe, chaoyu) have weighed-in consumption events logged to a ledger. A fourth participant’s ledger is populated with purchases only (annotation in progress); a fifth has started recording but has not yet begun annotation.

Table 1: Current collection progress vs. target.

	Current	Target
Participants (recording)	5	4
Participants (annotated ledger)	3	4
Total video	28.0 h	22 h
in-the-wild	27.5 h	20 h
controlled	0.4 h	2 h
Consumption events (weighed)	409	700
in-the-wild	377	400
controlled	32	300
Distinct tracked item instances	136	~70

Across the annotated ledgers, the 136 tracked item instances fall into four shape categories (Table 2 below breaks this down with examples): SOLID 39, DISCRETE 37, GRANULAR 33, FLUID 20, with 7 instances not yet categorised. In-the-wild hours already exceed the 20 h target, and the wild event count (377) is close to its 400-event target. The controlled cohort is the principal bottleneck — only ~0.4 h recorded and 32 events logged against targets of 2 h and 300 events — and is our near-term collection focus (see rationale in Section 3).

3.3 Item variety

Each item in the dataset is tagged with a *shape category* — one of {FLUID, GRANULAR, SOLID, DISCRETE} — along with a *transparency* flag indicating whether the package is see-through. These attributes are informative because they govern how an amount is estimated: fluid and granular items in transparent packaging permit fill-level reading, discrete items (eggs, yogurt cups) permit counting, and opaque containers must be inferred from pour/scoop duration or weighing context. Table 2 summarises the current item inventory.

Table 2: Item variety across all annotated ledgers, grouped by shape category. Counts are item instances (one row per purchase). Transparency labelling is ongoing.

Category	Items	Examples
Fluid	20	olive oil, milk, soy sauce
Granular	33	rice, pasta, cereal
Solid	39	garlic, steak, tofu
Discrete	37	eggs, yogurt cups, bagels
Unlabelled	7	(categorisation pending)
Total	136	

3.4 Longitudinal coverage

The same item often persists across many sessions before it is exhausted, which lets us evaluate *how a system tracks one package*

over its full lifecycle rather than only how it does on isolated sessions. Across the currently-annotated ledgers, item lifecycles span the distribution shown in Table 3: 32 item-instances appear in at least 5 distinct sessions, 11 in at least 10, and the longest-running items (eggs, sliced cheese, cereal, milk gallons) are observed across 13–18 sessions each. This long-tail subset is the substrate for the per-item lifecycle evaluation described in Section 4.5.

Table 3: Item lifecycle distribution across annotated ledgers. “Sessions” is the number of distinct sessions in which a given item instance was logged as consumed.

Sessions per item	Item instances	Cumulative ($\geq$)
≥ 1	92	92
≥ 3	55	55
≥ 5	32	32
≥ 10	11	11
≥ 15	3	3

The bulk of this longitudinal coverage currently comes from a single participant (kailai, 70 items). Bringing the other participants to comparable lifecycle depth, rather than to a wider item catalogue, is the principal collection focus alongside the controlled-cohort gap noted in Section 3.2.

4 BENCHMARK AND EVALUATION PLAN

4.1 Benchmark structure

Videos are organized into *sessions* (meal-prep episodes). For each session, the ground-truth label is, for every item touched, the *starting amount* and the *end amount*. The system under test takes a session video as input and produces a list of [item, remaining_amount] pairs as output.

4.2 Evaluation cohorts

We evaluate the two collection flavours for different purposes.

In-the-wild cooking videos. These are our primary cohort and the one that reflects the deployed use case. On this cohort we report *both* (i) amount-estimation accuracy (CNPE, Section 4.3) and (ii) VLM token consumption per session (Section 4.4). Target scale at submission: ~20 h of video and ~400 consumption events across 3 participants (numbers to be finalised).

Controlled consumption videos. We use the controlled cohort to evaluate VLM amount-estimation accuracy over a broader, deliberately diversified set of food items (see Section 3). The item mix here spans shape categories and packaging transparencies that are under-represented in any one participant’s natural cooking, which lets us measure how proposed system’s accuracy gain over baselines generalises across item types. Target scale: ~2 h of video and ~300 consumption events; we report CNPE on this cohort with paired per-event comparisons so that proposed system-vs.-baseline differences can be tested at statistical significance.

4.3 Primary metric: CNPE (accuracy)

We measure accuracy with **Capacity-Normalized Percentage Error (CNPE)**:

$$\text{CNPE} = \frac{|\text{prediction} - \text{GT}|}{W_0} \times 100,$$

where W_0 is the item’s original package size (capacity). Normalizing by W_0 makes errors comparable across items of very different sizes: a 50 g error on a 500 g jar (10% CNPE) is treated the same as a 50 g error on a 50 g packet (100% CNPE). CNPE can be computed equivalently on the remaining amount or the used amount.

4.4 Primary metric: VLM token consumption (cost)

We report total VLM token cost per session (input + output, with thinking tokens charged at output rate). Cost is treated on equal footing with CNPE: the central claim of the paper is a *cost–accuracy* trade-off, and a system that wins on one axis while losing on the other is not preferred unless the trade is explicitly characterized along the cost–accuracy frontier.

4.5 Cross-cut analyses

The two headline metrics (CNPE, token cost) are reported as session-level aggregates, but the questions that drive system design are sharper than a single mean. We additionally report three cross-cut breakdowns of the same underlying per-event predictions.

Per-class CNPE. CNPE is broken down by item shape category (FLUID, GRANULAR, SOLID, DISCRETE) and crossed with packaging transparency. The hypothesis is that the proposed system’s gain over the baseline concentrates on classes where amount reasoning is locally hard but *visible* given the right frame — fill levels in transparent fluids, residue in granulars, missing items in discrete counts — whereas opaque-fluid and irregular-solid estimation remain hard regardless of how the observation windows are chosen. Reporting per-class CNPE makes the source of any aggregate gain auditable and exposes packaging types where future work is most needed. The per-class event counts in Table 2 (20–39 instances per category, with multiple consumption events per instance) are sufficient to support per-class means with bootstrap intervals; the controlled-cohort expansion (Section 3.2) is targeted at the categories currently thinnest in event count.

Item-lifecycle CNPE. Inventory tracking is intrinsically a longitudinal task: the same package is observed across many sessions until exhausted. For the long-tail item subset characterised in Section 3.4 (32 item-instances with ≥ 5 sessions, 11 with ≥ 10), we report a per-item CNPE *trajectory* indexed by session number, alongside a single per-item lifecycle CNPE (mean across that item’s sessions). Two trajectory shapes matter and we will distinguish them: (i) session-by-session error that stays roughly flat, indicating the system re-anchors against the current frame and earlier session errors do not compound; and (ii) error that grows monotonically as the package empties, indicating that low-fill states are a systematic failure mode and motivating the cross-session visual memory extension (Section 2.1). For items observed all the way to a “finished”

event, we additionally report end-of-life cumulative drift (predicted total used minus actual, normalised by capacity).

4.6 Experiment matrix

Every cell in the matrix below is reported under both cross-cut breakdowns from Section 4.5 (per-class and item-lifecycle), so that the headline CNPE/token comparisons can be audited along the dimensions most likely to drive system design.

- **Controlled vs. natural sessions.** We expect amount-estimation error to be lower on controlled sessions (single item, scripted consumption) and will report the two cohorts separately.
- **Baseline vs. proposed system.** Whole-video baseline and the proposed system (planner + sweep observer with iterative replan per Section 2), using the same VLM backends for a fair comparison.
- **Ablations within the proposed system.** (i) Number of plan–observe rounds (1 vs. 2 vs. unbounded under a session-level frame budget); (ii) frame-budget allocation policy (journey-only, dense-only, mixed; late-burst-biased vs. uniform); (iii) planner evidence rendering (per-item temporal segments vs. chronological timeline vs. activity blocks); (iv) backbone VLM for the observer (GPT-5, Gemini 3 Pro / 2.5 Pro / Flash, Gemma) under matched frame-budget caps so the comparison isolates VLM choice from frame count.
- **Extension (Section 2.1).** We will additionally report *cross-session visual memory*: the proposed system with prior end-of-session snapshots threaded into the reasoner prompt, versus the same proposed system without — evaluated specifically on the long-tail items of Table 3 where the trajectory metric (Section 4.5) makes any cross-session benefit visible. The extension is evaluated on the same CNPE/token axes so its cost/accuracy trade-off against the base proposed system is directly comparable.
- **Scalability.** As more participants and sessions come online, we will re-run the full matrix and report metric stability as the experiment set grows.

4.7 Locked evaluation definitions

To make the small-scale pilots and the final-cohort runs directly comparable, three definitions are pinned for the remainder of the project. Any later change to these is treated as a methodological revision and the affected runs are re-executed.

Frozen cohort lists. The wild and controlled cohorts are specified as explicit session-ID lists per participant (Appendix ??, to be added). New sessions recorded after the freeze date are excluded from the headline numbers and may appear only as out-of-distribution diagnostics. The current pilot cohort is the 10-session canonical kailai natural-cooking subset; the final cohort substitutes the larger frozen list under the same protocol.

Frozen pricing convention. Billable cost per session is reported in USD using Gemini 3 Pro paid-tier pricing (\$1.25/M input, \$10/M output, thinking tokens charged at output rate), with images converted at 1 frame = 260 input tokens. The same pricing convention is applied to the whole-video baseline and the proposed system so that cost comparisons are not confounded by per-comparator pricing

choices. Planner LLM cost is included only in the proposed system and extension rows.

Frozen metric definitions. Three metrics constitute the headline:

- **CNPE per event** (Section 4.3), reported as the mean over GT consumption events.
- **Recall** = the fraction of GT consumption events for which the system produced a non-null amount estimate. Items the planner explicitly skipped, items the observer marked as not-confirmed, and items never proposed in the first place all count as missed.
- **Fused CNPE+recall** = the per-event mean of CNPE with missed events charged at CNPE = 100. This single scalar is the y-axis of the headline cost-accuracy figure. The failure floor of 100 corresponds to a full-package error and prevents systems from improving the headline by skipping uncertain items.

5 RESULTS

This section is a skeleton. Each figure and table below is defined here so that experiments executed between now and the freeze date populate specific cells rather than producing orphaned runs. Figures and tables will be filled as experiments complete.

5.1 Setup

Final cohort sizes (sessions, hours, consumption events) per participant, broken out by wild vs. controlled. One row per comparator (whole-video baseline, proposed system R1, proposed system R1+R2, proposed system+memory) listing the locked run configuration: planner LLM, observer VLM, frame-budget cap, evidence rendering mode, burst-coverage rule, thinking depth. Pricing convention restated from Section 4.7.

5.2 Headline: cost-accuracy frontier

Figure (central): Pareto plot. x-axis = USD/session (log scale); y-axis = fused CNPE+recall (lower is better). One marker per (comparator × frame-budget) cell. Curves: whole-video baseline (single point), proposed system frontier (sweep over frame budget and round count). Iso-cost guides at \$0.10, \$0.50, \$1.00. Extension points appended in red when implemented.

Table 4 (placeholder): aggregate CNPE, recall, and \$/session for each comparator on the wild and controlled cohorts.

5.3 Cross-cut analyses

Two panels populate the breakdowns specified in Section 4.5:

- **Per-class CNPE bars:** shape category (FLUID/GRANULAR/SOLID/DISCRETE) × comparator, crossed with packaging transparency. Tests the “gain concentrates on visible-but-locally-hard classes” hypothesis from Section 4.5.
- **Lifecycle trajectory grid:** small multiples, one panel per long-tail item from Table 3, x = session index, y = CNPE, two lines per panel (whole-video baseline / proposed system). End-of-life cumulative drift reported in an adjacent table for items observed to “finished”.

5.4 Ablations

Each ablation produces a small Pareto subplot sharing axes with the headline figure. (i) Round count (R1 vs. R1+R2 vs. unbounded under a session-level frame budget). (ii) Frame budget allocation policy (journey-only / dense-only / mixed; late-burst vs. uniform). (iii) Planner evidence rendering (blocks vs. chrono vs. segments). (iv) Observer-backbone swap (GPT-5, Gemini 3 Pro, Gemini 2.5 Pro/Flash, Gemma) at matched frame-budget caps.

5.5 Failure analysis

2–3 hand-picked sessions with figure-strip illustrations: session timeline, frames the observer actually saw, predicted vs. true amount, one-sentence diagnosis. Each case is mapped to either the cross-session memory extension or flagged as an open problem.

5.6 Extension

Cross-session visual memory results: matched-frame-budget comparison against base proposed system, evaluated on the Table 3 long-tail items so the lifecycle trajectories from Section 5.3 can show the benefit.

Table 4: (Placeholder.) Headline CNPE, recall, and \$/session per comparator on the frozen cohorts.

Comparator	Wild			Controlled		
	CNPE	Recall	\$/sess	CNPE	Recall	\$/sess
Whole-video baseline	–	–	–	–	–	–
proposed system (R1)	–	–	–	–	–	–
proposed system (R1+R2)	–	–	–	–	–	–
proposed system+mem	–	–	–	–	–	–

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